TrES-5b

R-0776

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_231031` · created 2026-07-28T05:41:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460248.6929 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.037
Transit depth (Rp/Rs)^2	1.17 +/- 0.8 [%]
Orbital Inclination (inc)	89.61 +/- 2.69
Airmass coefficient 1 (a1)	1.18 +/- 0.029
Airmass coefficient 2 (a2)	-0.1 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.42 %
Best Comparison Star	#3 - [272, 327]
Optimal Method	PSF photometry
Transit Duration (day)	0.0818 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.108 ± 0.037 vs 0.142 (deviation 0.85σ)
Duration fit vs published	0.0818 d vs 0.0758 d (deviation 0.9σ)
Mid-time O–C	-5.5 min
Depth SNR	2.7
Residual scatter	2.42 %

Light curve

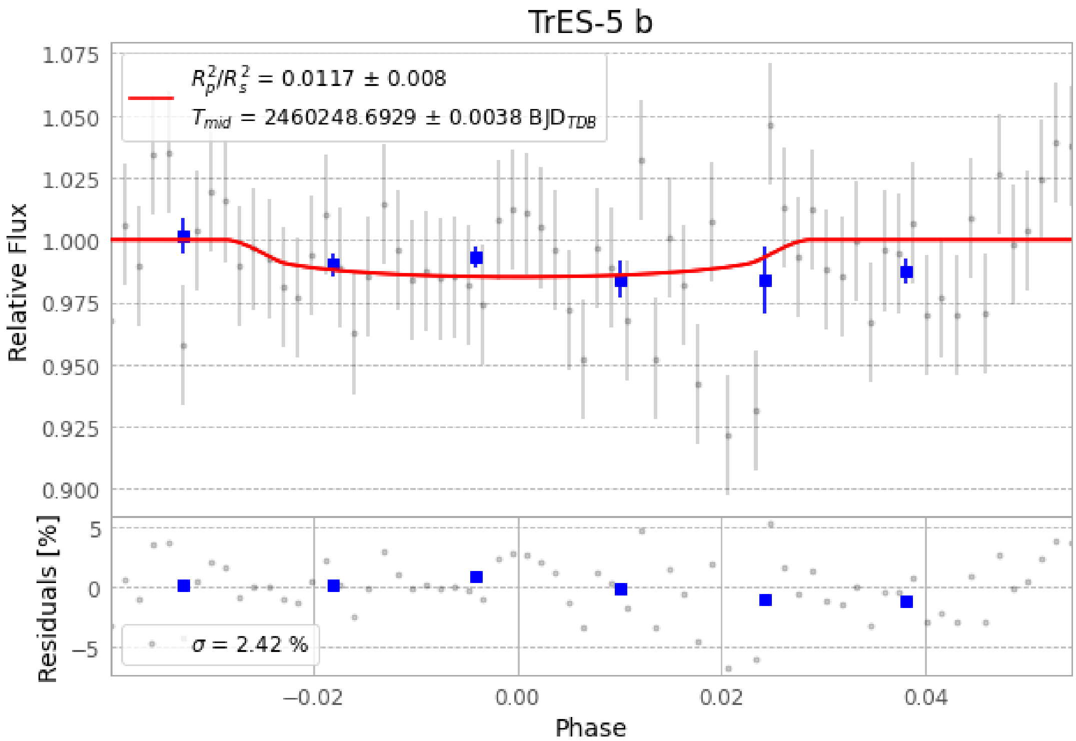

Figure 1 — FinalLightCurve_TrES-5 b_2023-10-30.png (SHA-256 4114b2940482ca56…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 188], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 83d4d102b05a09df…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

68 FITS frames (45 MB), `TRES-5231031030917.FITS` → `TRES-5231031063015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-231031.log (2a794aa14344…), FinalLightCurve_TrES-5 b_2023-10-30.png (4114b2940482…), FinalLightCurve_TrES-5 b_2023-10-30.csv (db5dc2553ca1…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)